



7495 Shovels - Testing and Assessing Condition of the IGBT Skid DC Bus Capacitor Bank, as well as Discharge and Ground Tie Resistors

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

The purpose of this document is to provide methodology to test, and assess the condition of DC Bus (DC Link) Capacitors, DC Bus Precharge/Discharge Resistors, and the Ground Tie (Ground Fault) Resistors on board the 7495 shovels.

DC Bus Capacitors are utilized in the IGBT Skid (Motion Drive) to stabilize and store DC Current produced by the AFE (Active Front End) IGBT's while working in the Motoring (Rectification) Mode.

This current is then used by the individual Motion Inverters (Hoist, Swing and Crowd) to produce Variable Frequency AC Current, required for fluent motor speed and torque control:



ELECTROCUTION HAZARD - The electrical system contains hazardous voltage levels.

DO NOT remove any covers that will expose energized hazardous-voltage electrical components without shutting down and discharging the electrical system.

DO NOT perform any maintenance on the electrical system until after the electrical system has been shut down and the voltage has been discharged by qualified personnel.

Read and understand the instructions and warnings found in the Operation and Maintenance Manual to properly shut down the electrical system and discharge hazardous-voltage electrical components.

Failure to follow these instructions will result in personal injury or death.



When service or maintenance must be performed in a hazardous-voltage component enclosure, always assume that a hazardous voltage level could be present.

Qualified personnel should manually check for hazardous voltage using electrical measuring tools that have the capacity to measure the hazardous-voltage components being checked.

The voltage level present must be confirmed to be below 50 Volts before any exposure to the hazardous-voltage components can take place.

Failure to follow these instructions could result in personal injury or death.

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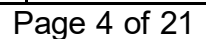


Fig 1 – Location of DC Bus (DC Link) Capacitors in the 7495 IGBT Skid – typical



Fig 2 – DC Bus Capacitors C12, C22, C13 & C23 in the Crowd Inverter Cabinet (left), and Capacitors C32, C42, C33 & C43 in the Swing Inverter Cabinet (right)

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Although the DC Bus (DC Link) Capacitors are very reliable components, they may fail due to temperature extremes (overheating), voltage spikes, but most commonly due to electrical erosion (arcing) at the Power Connecting Blades or Sockets:

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Fig 4 – Capacitor Power Connecting Blade deteriorated by arcing – typical



Fig 5 – Capacitor Backplane Connections (Sockets)

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3.0 Implementation Steps

Prior to inspecting any electrical components connected to the DC Bus, perform the mine approved machine isolation procedure and ensure that the DC Bus has been discharged. Refer to **Special Instruction M0071820-01**

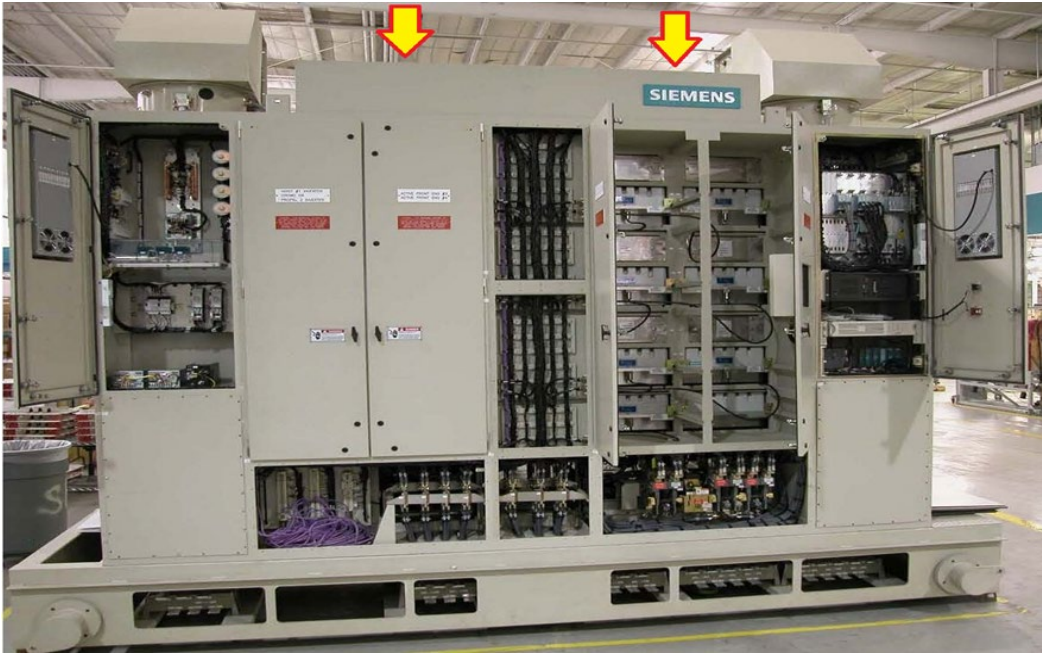


Fig 7 – Location of Discharge & Pre-charge Resistors atop the IGBT Skid



Fig 8 – View of DC Bus Discharge and Ground Fault Resistors – left side facing Skid

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Fig 9 – View of DC Bus Discharge Resistors and Pre-charge Limiting Resistor – right side of Skid

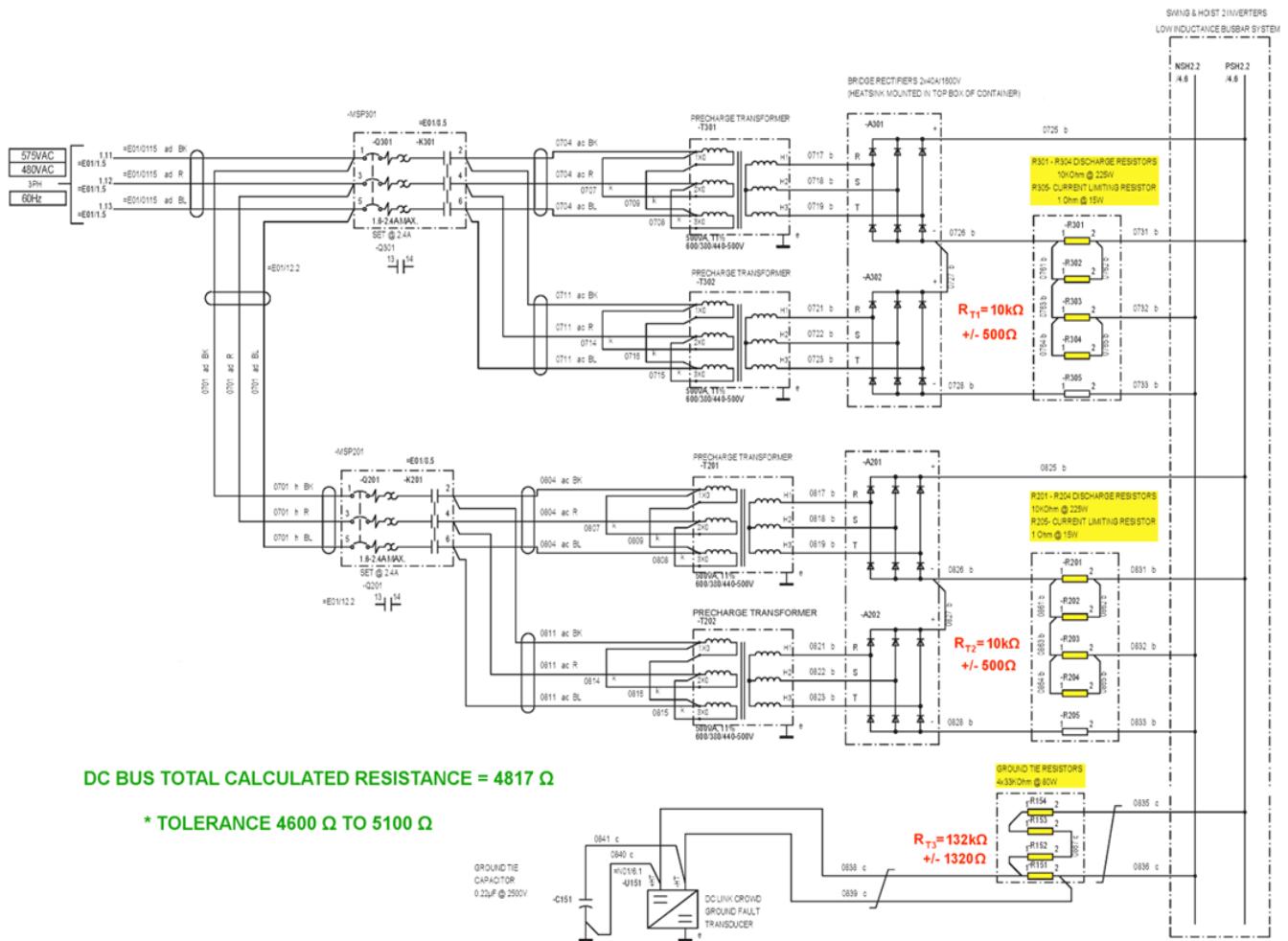


Fig 10 – Total resistance values of DC Bus and each Resistor section

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In order to quickly verify whether or not the DC Bus Pre-charge/Discharge Resistors and the Ground Tie Resistors did not deteriorate with time and operating conditions, it is recommended to obtain quick reference reading of DC Bus Resistance, should the actual total DC Bus Resistance be significantly different then calculated value of **4817 Ω +/- 300 Ω**

To perform this task, ensure that all applicable safety procedures have been applied, and that the DC Bus has been discharged.

At this point, locate the DC Bus (DC Link) Voltage Transducers U152 (wires 0842 & 0843) and U452 (wires 0742 & 0743), and use a quality multimeter set to Ohm's scale, to read the DC Bus Total Resistance. Record this value, to be used later in Capacitor Test Calculator.

Refer to images below for reference:

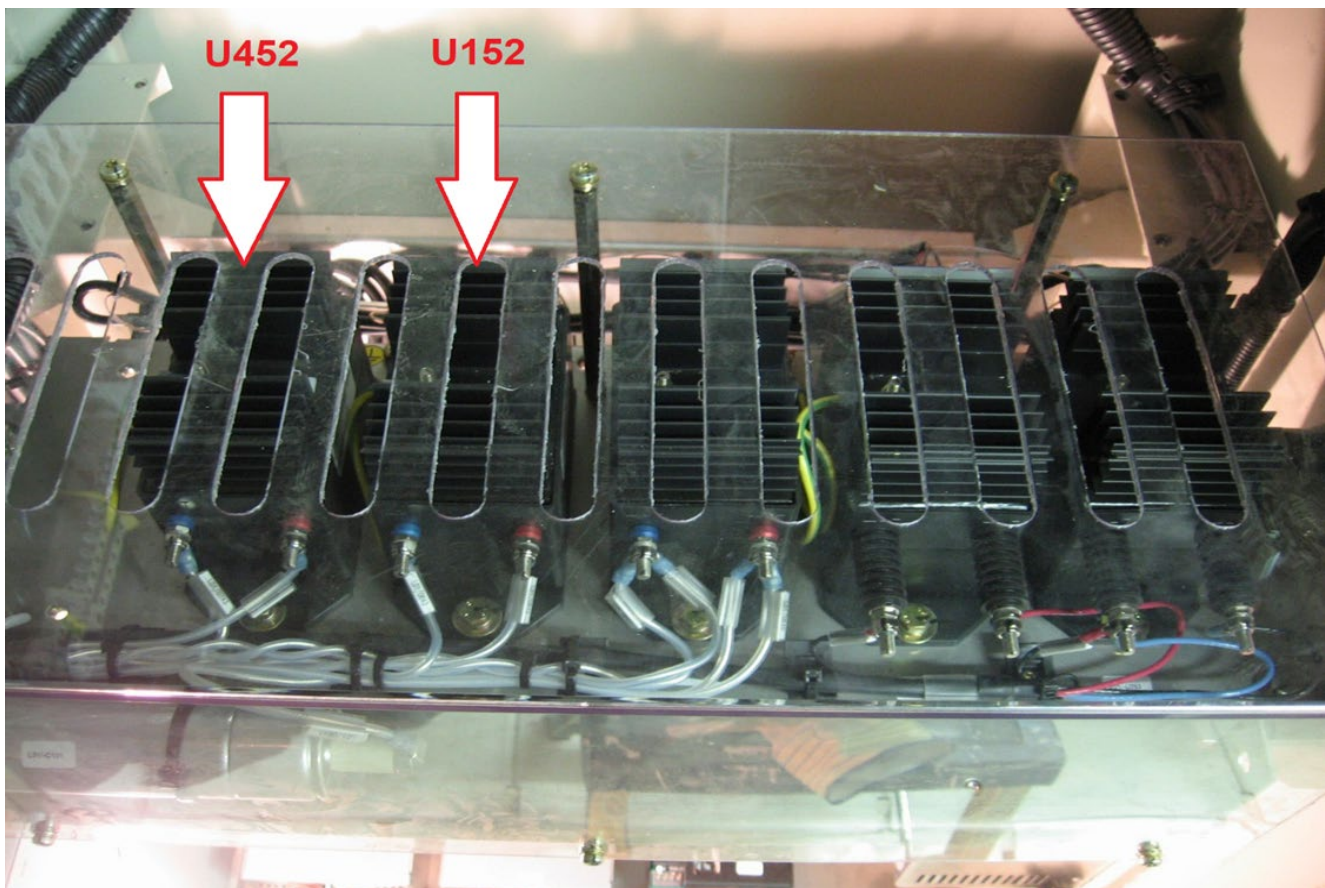


Fig 11 – Location of DC Bus (DC Link) Voltage Transducers U452 and U152 in the Transducer Tray

NOTE: This Transducer Tray is in the Control / Power Supply Cabinet of the IGBT Skid on the older vintage machines, and in the cabinet located at the back of the SIBAS Cabinet (rear of skid) on the newer machines.

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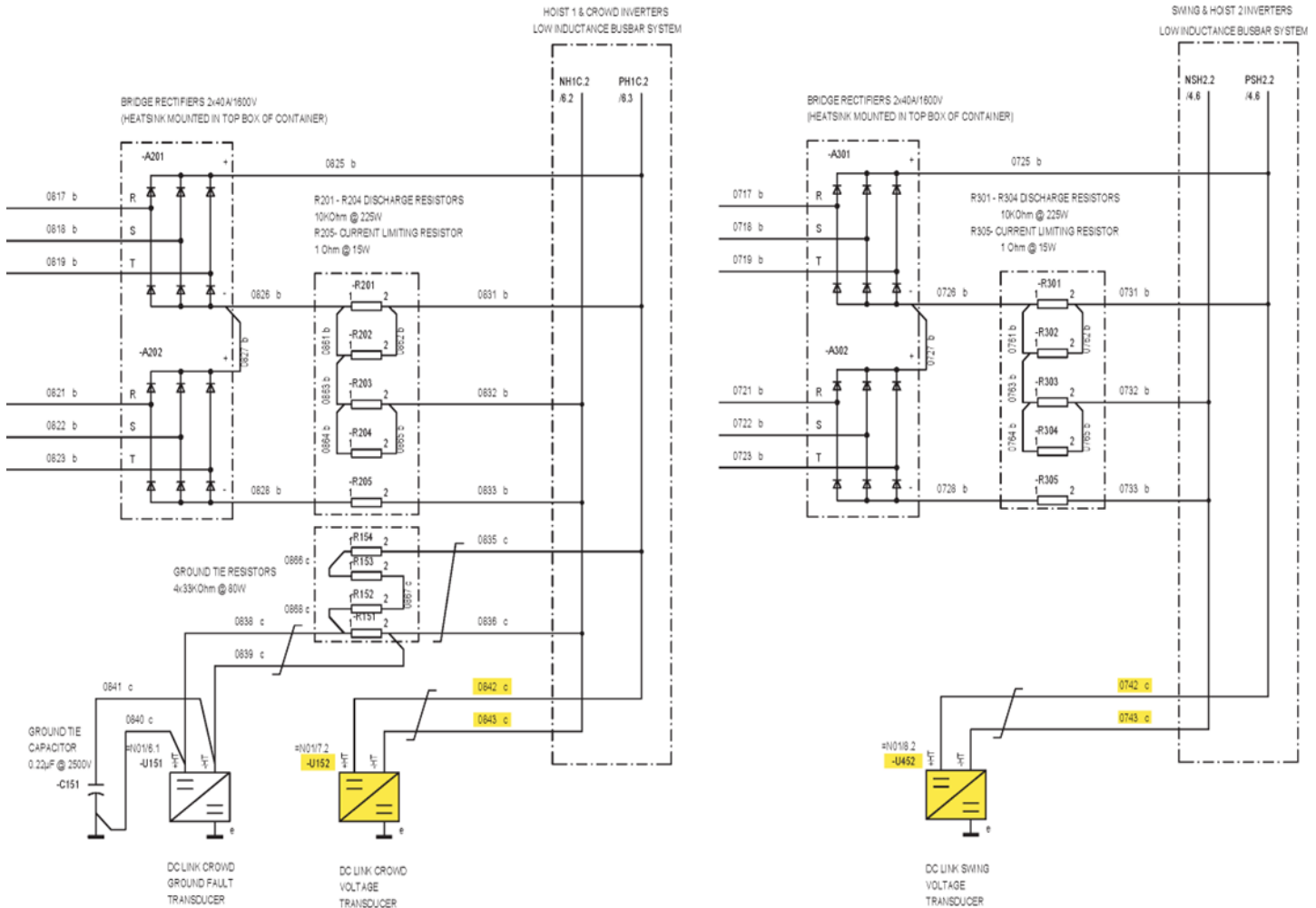


Fig 12 – Schematic of U452 and U152 Connections to DC Bus

In order to test and assess the condition of DC Bus Capacitor, a (Windaq/Dataq) recording will be required, followed by review of the recordings, and entry of captured data into specially prepared Calculation Sheet.

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Step 1] Using the Reference Recording provided with this publication, open the WinDaq Pro+ Data Acquisition software and create a recording of the required signals.

NOTE:

Please verify that the Test Panel channels are set to +/- 10V FS (Full Scale), and the BEB Circuit Channel is set to +/- 1000V FS to avoid potential oversaturation and damage to DATAQ Recorder Channel Amplifiers

Step 2] Using the IGBT Skid Test Panel, as well as Siemens Schematics, connect the DATAQ (Windaq) Recorder to capture the following signals:

- a) DC Bus Voltage Transducers (DCV#1 and DCV#2)
- b) Chopper Current Transducers (ICH#1 and ICH#2)
- c) BEB Pushbutton

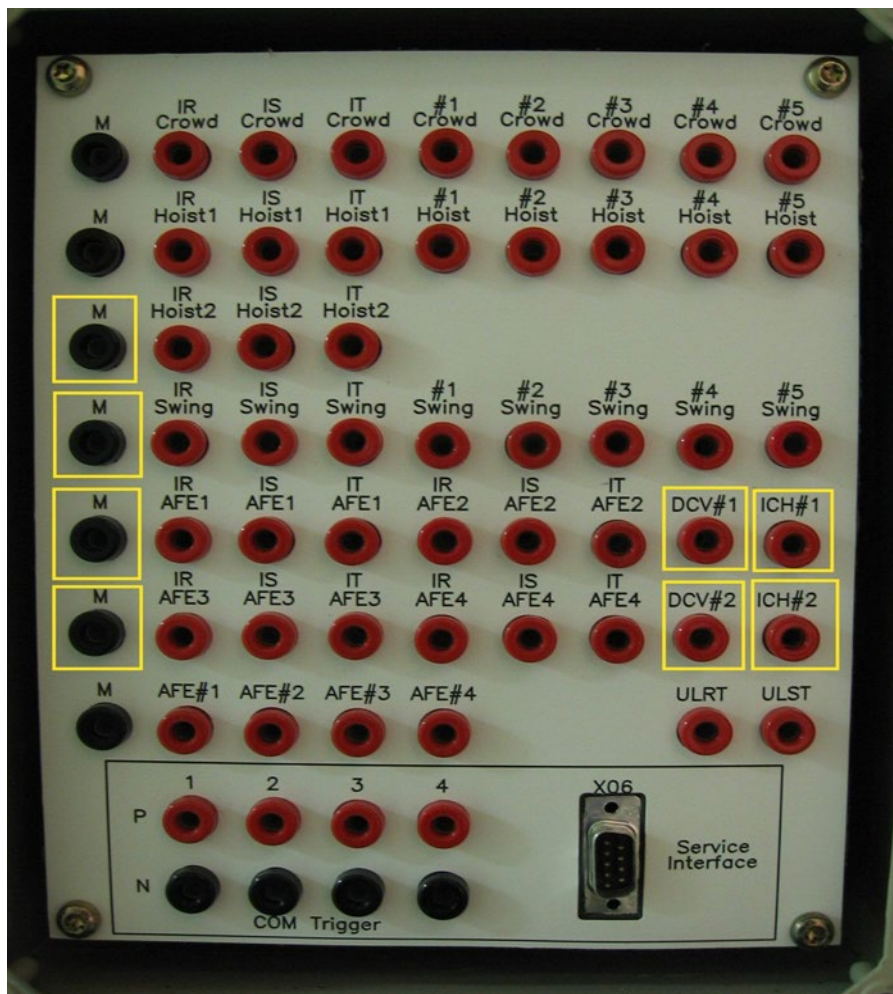


Fig 13 – Test Panel recording points

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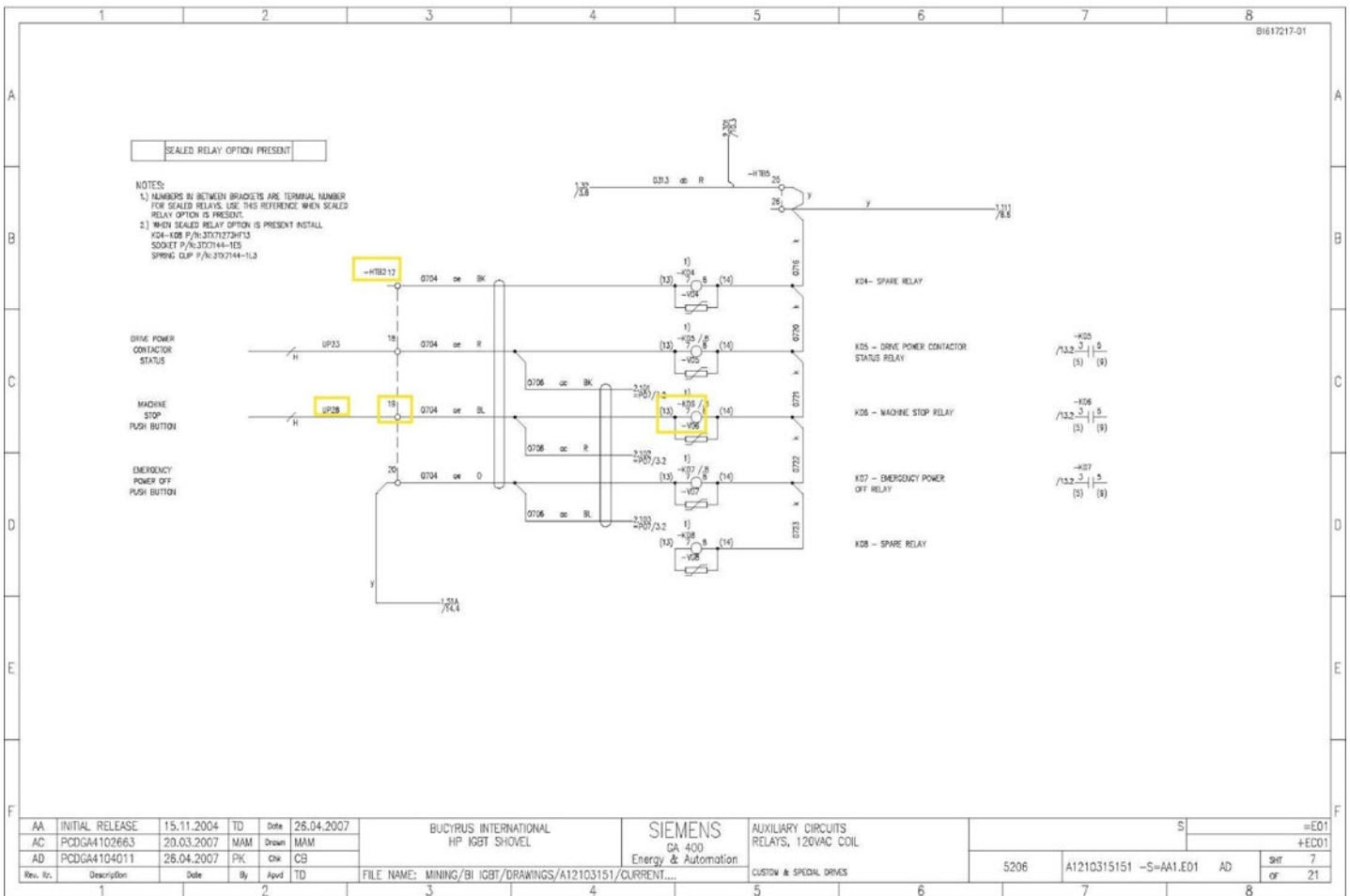


Fig 14 – BEB Circuit (wire UP28) HTB2-19 or K06 recording point

Warning: This is a 120V AC circuit, and connecting the Dataq / Windaq recorder may require removal of power by turning off the UPCB Breaker in the ZAC Cabinet

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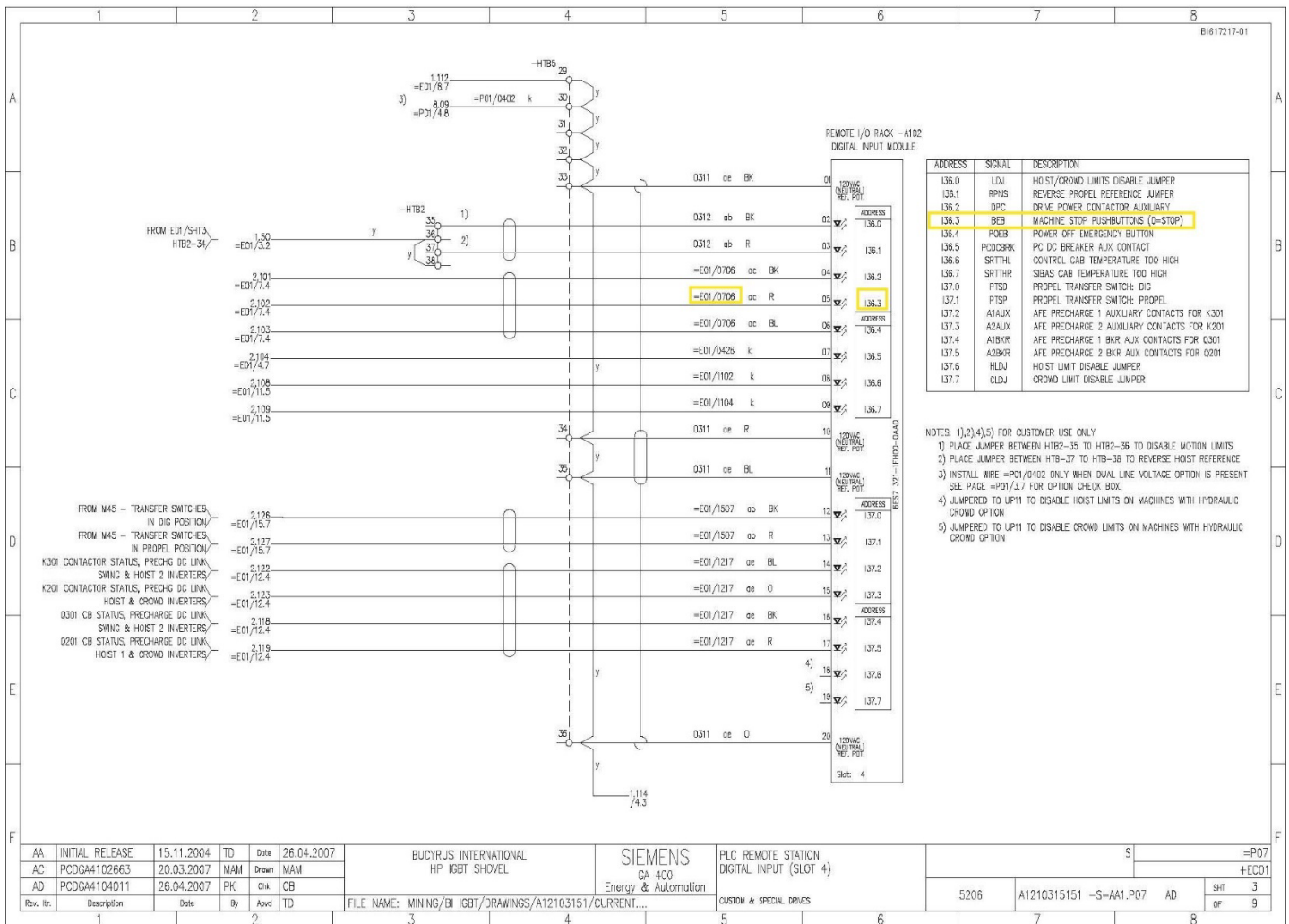


Fig 15 – BEB Circuit NODE 7 PLC (Input I36.3) alternative recording point

Warning: This is a 120V AC circuit, and connecting the Dataq / Windaq recorder may require removal of power by turning off the UPCB Breaker in the ZAC Cabinet

STEP 3] If the IGBT Skid is still running (DC Bus is up and approximately 1800V is displayed on the HMI overview screen), start the Recorder, and then push the **BEB** button.

If the shovel is not running, start it up, allow the DC Bus to reach 1800V DC and stabilize (Allow to run for ~1 minute after reaching 1800V), start the Recorder, and then push the **BEB** button.

Shortly after the DC Bus value on the HMI Overview Screen reaches “zero “volts, stop the Recorder, and close the recording.

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STEP 4] After the recording has been stopped, utilize the WinDaq Waveform Browser to replay the Recording.
Since we will be investigating the DC Link Voltage, as well as elapsed time since BEB has been activated, using the **View** dropdown menu, format the screen to display only the three significant channels / signals of interest, utilizing the **User-specified format** option:

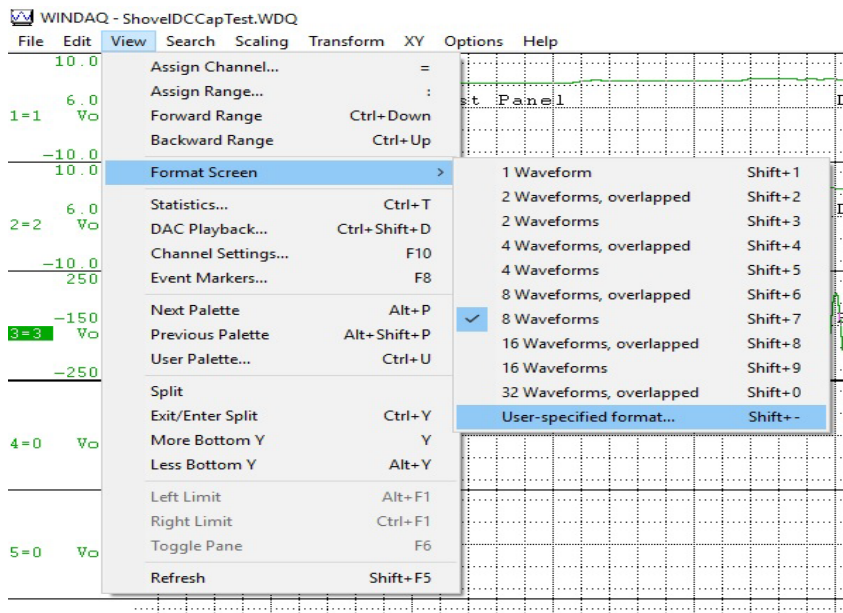


Fig 16 – Formatting Screen to User-specified format

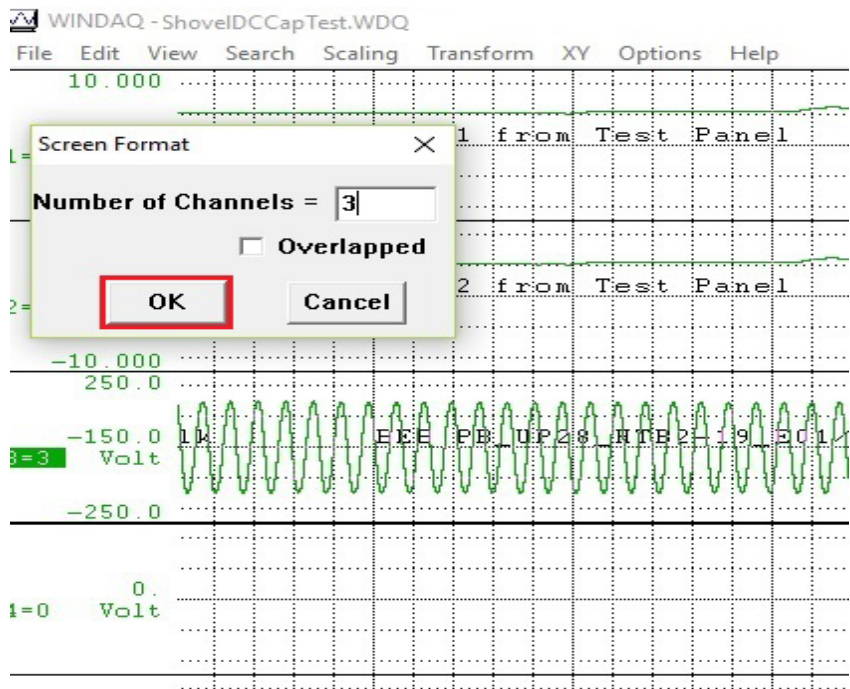


Fig 17 – Formatting Screen to display 3 top channels only

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STEP 5] At this point, use the **Compression** option in the **Edit** dropdown menu, to find the point on the recording where the BEB Pushbutton was activated. In this example compression of 20 was used:

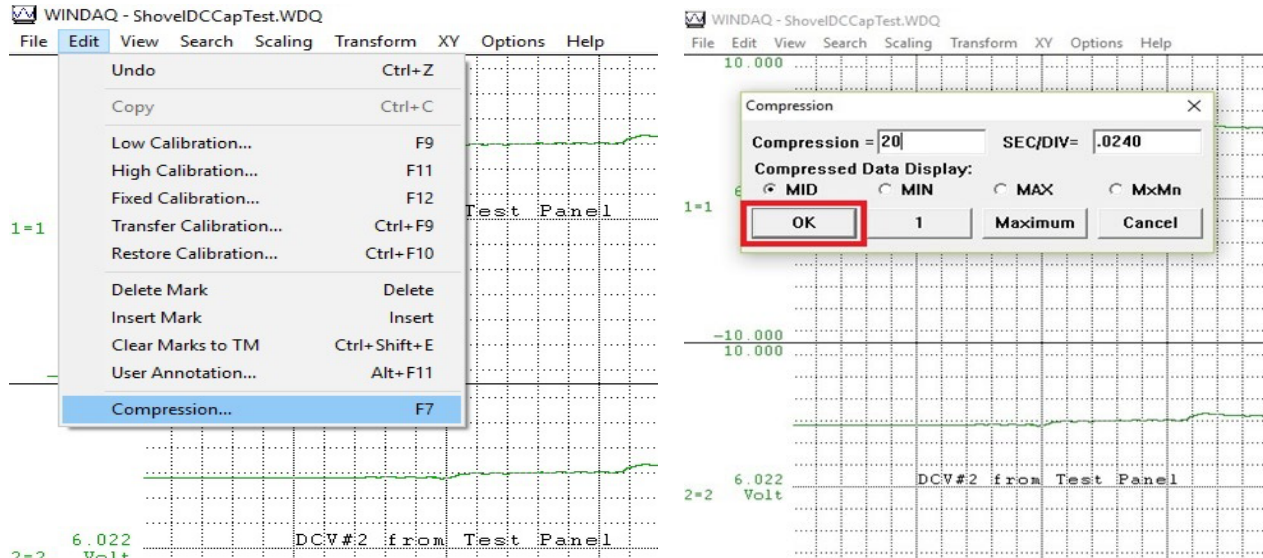


Fig 18 – Applying compression factor of 20

Once the compressed screen appears, move the Cursor Line by clicking the mouse immediately below the Cursor Line, and dragging it to the beginning of area where the BEB signal dropped off.

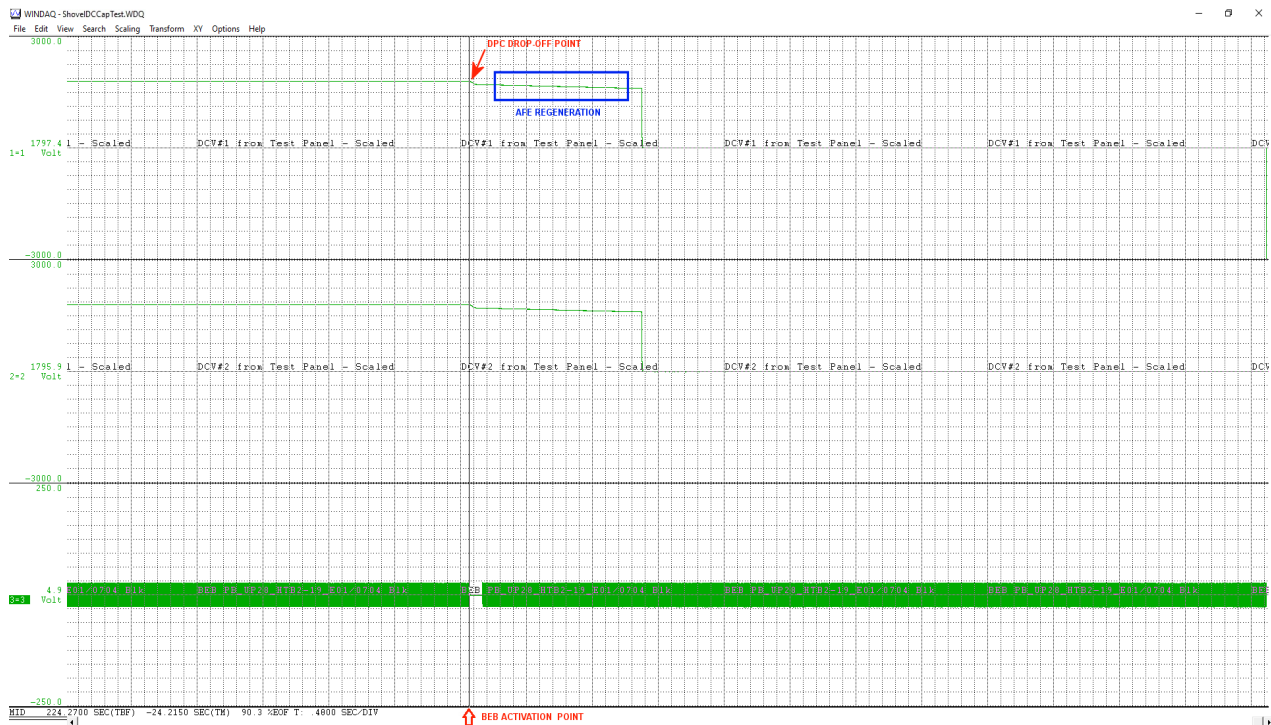


Fig 19 – Moving the Cursor Line to the point of interest

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Now observe the AFE Regeneration Points (dips) on the recording and move the cursor to the point passed the end of the last regeneration point. At this stage, decompress the recording to compression factor of 1.

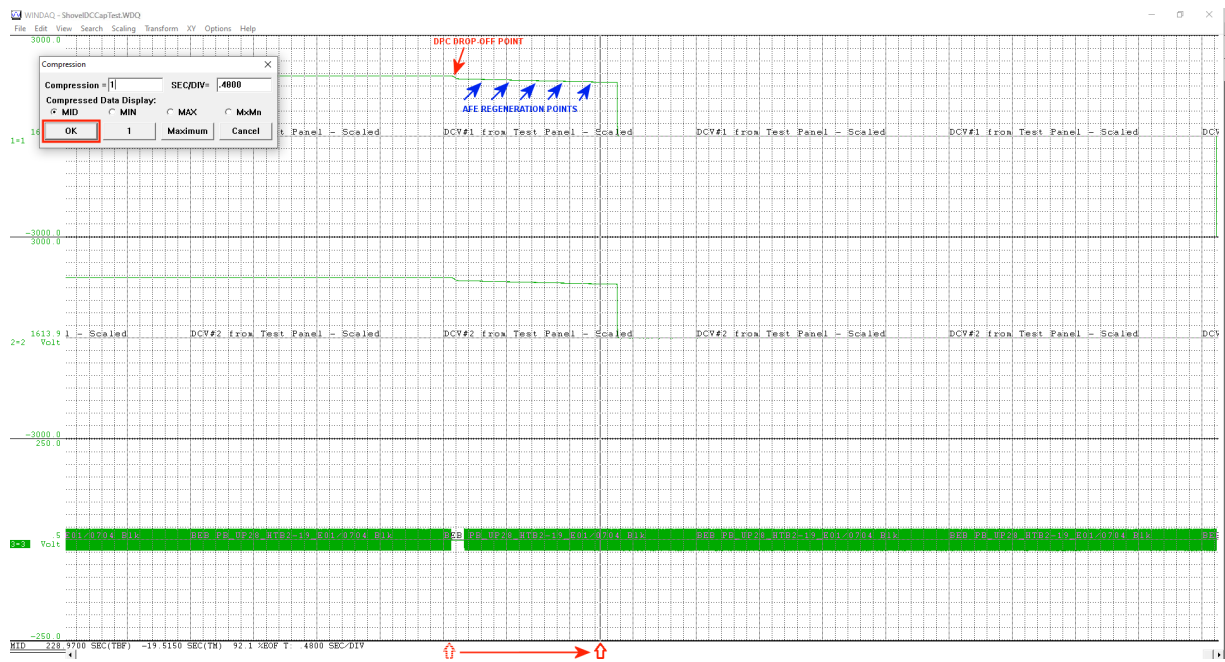


Fig 20 – Relocating cursor and decompressing the recording to original state – compression of 1

STEP 6] At this point, review the DC Link Voltage signal to find the area of the last AFE Regeneration Point, and move the Cursor Line to the edge of the very next division (vertical grid line) on the recording. This will be the starting point of the voltage and time measurement:

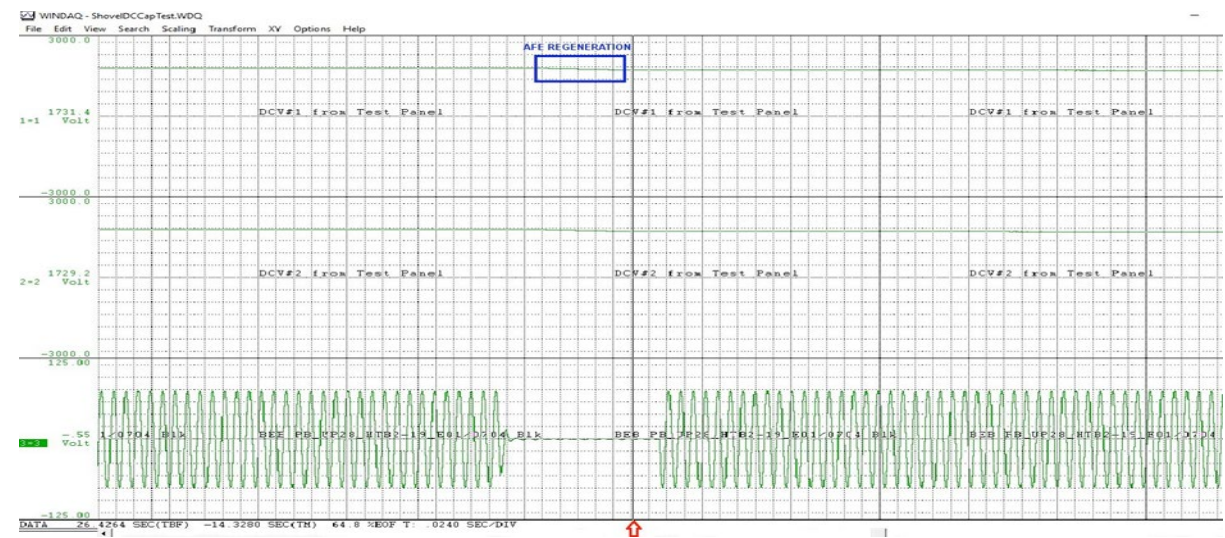


Fig 21 – Moving Cursor Line to the next division just passed the AFE Regeneration Point

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STEP 7] From the position established as the **Starting Point of the measurement**, make note of the DC Bus Voltage and Recording Time:

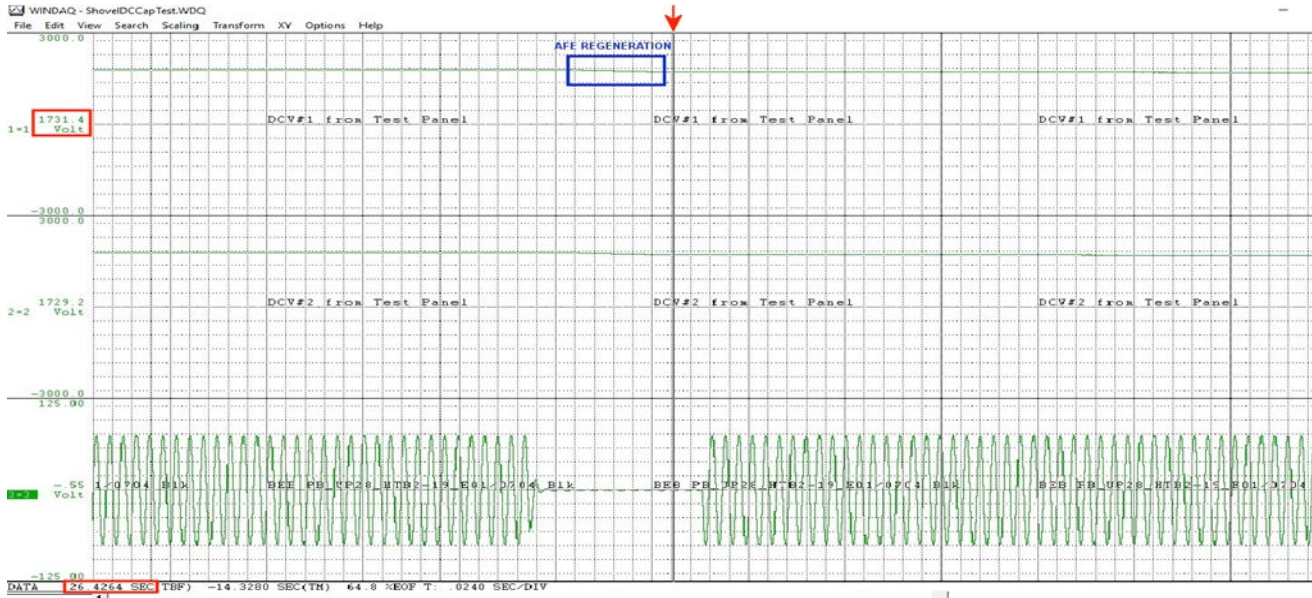


Fig 22 – Starting point of the measurement - note the DC Bus Voltage and Recording Time

STEP 8] Now move the recording forward (to the right) by using the right Arrow on the Slider Bar, or the Slider button at the bottom of the screen, to the division line just prior to the voltage signal dropping off (Choppers firing point). From the position established as **End Point of the measurement**, make note of the DC Bus Voltage and Recording Time:

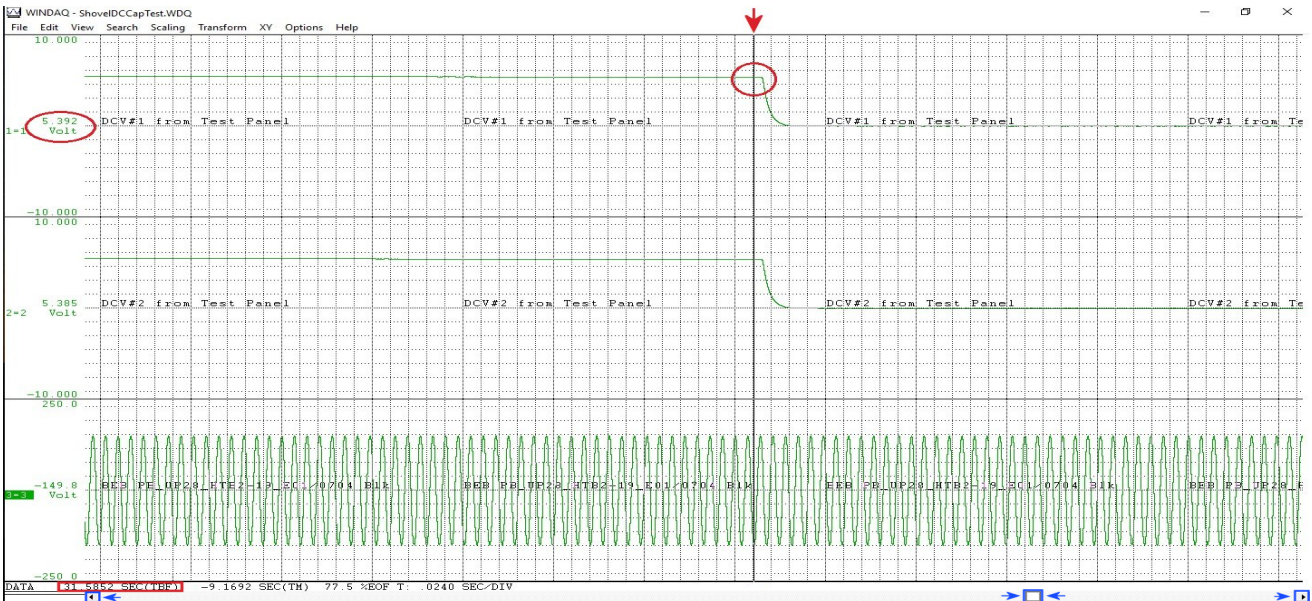


Fig 23 – End point of the measurement - note the DC Bus Voltage and Recording Time

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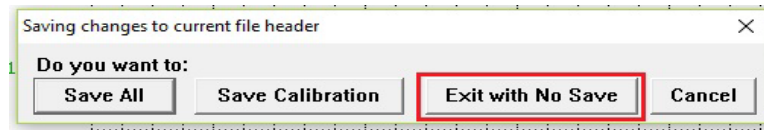
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STEP 9] At this point close the WINDAQ Software opting for “ Exit with No save “ ,



and utilize the **7495 IGBT Capacitor Test Calculator** provided with this publication, to enter values previously noted at the **Starting point of the measurement**, the **End point of the measurement**, as well as the **DC Bus Voltage and Resistance** into the Excel spreadsheet:

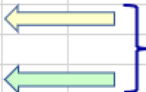
	A	B	C	D	E	F	G	H
1	IGBT CAPACITOR CHECK CALCULATOR							
2								
3	Input the voltages and elapsed time from Windaq recording:				* If DC Bus Resistance has not been measured, use Calculated value			
4								
5	Enter Initial Voltage (following BEB - passed the AFE regeneration points)	1731.4	Enter	Total Bus Resistance Measured		4817	Ohms	
6	Enter Start of Measurement Time Stamp	26.4264						
7	Enter End of Measurement Time Stamp	31.5852	Total Bus Capacitance Calculated =		0.016	Farads		
8	Calculated Elapsed Time	5.1588						
9			Total Bus Resistance Calculated =		4817	Ohms		
10								
11								
12	Enter End of Measurement Voltage (prior to Choppers firing) - Value X	1617.6			ACCEPTABLE RANGE +/- 6 VOLTS FROM CALCULATED VALUE - Value Y			
13								
14	Calculated Voltage with full capacitance - Value Y	1619.303						
15	Calculated Voltage with 1 bad module	1603.892	loss of 1 Cap. results in ~ 16V drop from Value Y					
16	Calculated Voltage with 2 bad modules	1583.573	loss of 2 Cap. results in ~ 36V drop from Value Y					
17	Calculated Voltage with 3 bad modules	1555.559	loss of 3 Cap. results in ~ 64V drop from Value Y					
18	Calculated Voltage with 4 bad modules	1514.463	loss of 4 Cap. results in ~ 105V drop from Value Y					
19	Calculated Voltage with 5 bad modules	1448.368	loss of 5 Cap. results in ~ 171V drop from Value Y					
20	Calculated Voltage with 6 bad modules	1324.707	loss of 6 Cap. results in ~ 295V drop from Value Y					
21	Calculated Voltage with 7 bad modules	1013.543	loss of 7 Cap. results in ~ 606V drop from Value Y					
22								
23	* NOTE: INPUT THE RECORDING VALUES ONLY IN THE AREAS HIGHLIGHTED IN YELLOW AND ANNOTATED WITH THE COMMAND " Enter "							
24	USER NOTES:							
25								
26								
27								
28								
29								

Fig 24 – IGBT Capacitor Check Calculator with recording values entered - full capacitance present

Important Note:

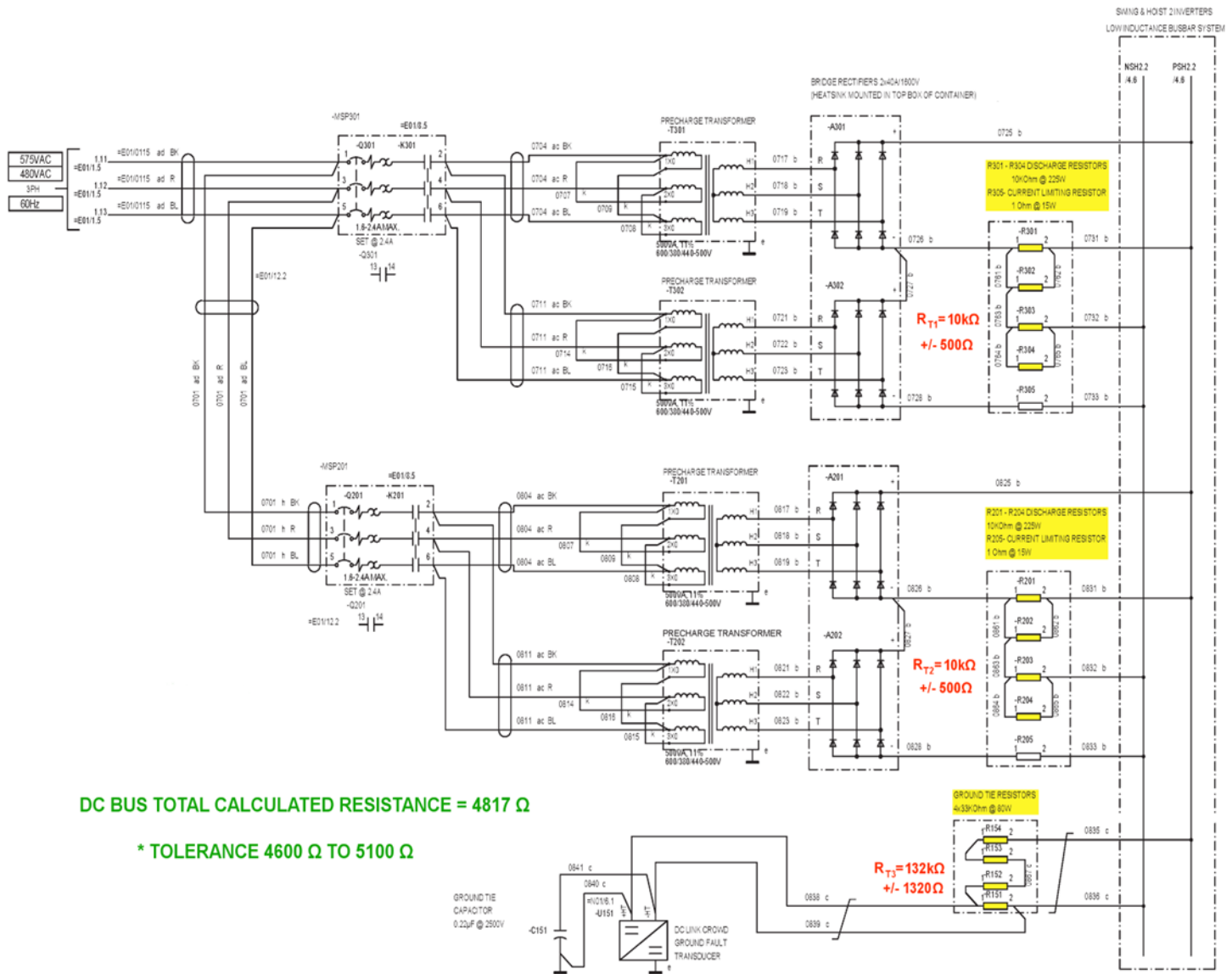
Please enter the relevant values obtained from the recordings in the cells highlighted in yellow, in the rows annotated with the word “Enter” only.
Do not attempt to change any other values, as this will falsify the calculation.

If the measured DC Bus Resistance was different from the calculated value of 4817 Ω +/- 300 Ω , then enter the Total Bus Resistance Measured value in the IGBT Capacitor Check Calculator.

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STEP 10] Should the measured DC Bus Resistance be different from the calculated value of 4817 Ω +/- 300 Ω , at this point, inspect condition of Resistors, wiring and terminations atop the IGBT Skid. Removal of IGBT Skid Top covers will be required – for location reference see Fig 7 through Fig 9, on pages 8 and 9 of this publication.

Disconnect one wire feeding each Resistor Section, and using an Ohmmeter verify resistance against the annotated drawing from **Fig. 10** below:



The acquired resistance readings, and comments resulting from the inspection can be entered in the **7495 IGBT Capacitor Test Calculator**.

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4.0 Benefits

Early detection of DC Bus capacitance fluctuation will ensure solid power source for the IGBT Modules.

It has been proven through historical machine failure data that failure of DC Link Capacitor will consequently lead to failure of the IGBT Module, and thus testing of the DC Link Capacitors should become a part of annual preventive maintenance practices.

Implementation of this practice will increase machine reliability, and control Total Operating Costs of the 7495 Shovel.

5.0 Resources Required

- High Voltage Gloves and Capacitor Discharge Tool as per Special Instruction M0071820-01
- 8 Channel Analog Input DATAQ Recorder
- WinDaq Pro+ Data Acquisition software, and WinDaq Waveform Browser software
- Laptop preloaded with the above WinDaq Acquisition and Browser software
- BNC Test Cables with Banana Plugs for the use on Test Panel Points (+/- 10V DC)
- Test Leads with Banana Plugs for BEB Circuit leads (120V AC) rated at 1000V DC.



Fig 25 –Twisted pair of 1000V Test Leads (left) and BNC low voltage cable (right)

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6.0 Supporting Attachments / References

Attachments:

- 7495 IGBT Capacitor Test Calculator
- Special Instruction M0071820-01
Procedure to Verify No Power on the IGBT in DC Bus
- Disassembly & Assembly Procedure M0069644-50
DC Bus Capacitor – Remove & Install (including discharge information)
- Service Magazine M0078294-01
Inspection Procedure for DC Bus Capacitor
- DC Bus Discharge Reference Recording – Master
- Instructions how to use WinDaq Reference File (Recording)

References:

Parts Reference:

DC Link Discharge Resistors 10k Ω	BI 62041410	SAP 1125430	CAT 472-3609
Ground Tie Resistors 33 k Ω	BI 62041411	SAP 1125431	CAT 472-3610
DC Bus Capacitor 2000 μ F	BI 62039652	SAP 1124165	CAT 472-6699
Capacitor & IGBT Power Tips Grease	BI 1579910	SAP 1579910	CAT 529-3659

7.0 Related Best Practices

Not available

8.0 Acknowledgements

SPECIAL THANKS to Caterpillar's Electrical Service Manager, Electric Rope Shovels, PLC, Drive System, for his engineering assistance in creating this publication.

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